



wp: Counterfactuals

Q1. (a) $V_M(\phi \Rightarrow \psi, u) = 1$ iff $V_M(\phi, u) = 0$ for all $u \in W$ or $V_M(\psi, v) = 1$ where v is the closest ϕ -world to u

$$\text{iff } V_M(\phi, u) = 0 \text{ for all } u \in W \text{ or } V_M(\psi, f(\phi, u)) = 1$$

The claim reduces to: Show we can define $f(\phi, u)$ s.t.

How do you know there is only one such v ?

$$V_M(\psi, v) = 1 \text{ for a maximally close } \phi\text{-world to } u, v \\ \text{iff } V_M(\psi, f(\phi, u)) = 1$$

$$\text{We can define } f(\phi, u) = \begin{cases} v & \text{if } V(\phi, u) = 1 \text{ and } \forall u \in W \text{ s.t. } V_M(\phi, u) = 1, v \leq_u u \\ \text{undefined} & \text{if } V_M(\phi, u) = 0, \forall u \in W \end{cases}$$

I don't think you should leave it undefined, because then

Lemma: v is a maximally close ϕ -world to u iff $f(\phi, u) = v$

Pf. ($\Rightarrow$) Assume v is a maximally close ϕ -world to u

$V_M(\psi, f(\phi, u)) = 1$ does not make sense w/ $V_M(\psi, x) = 0$ for all $x \in W$

$$\therefore v \text{ is a } \phi\text{-world} \Rightarrow V_M(\phi, v) = 1$$

and

$v \leq_u u$ for any ϕ -world u i.e.

$$\forall u \in W \text{ s.t. } V_M(\phi, u) = 1, v \leq_u u$$

which is just the definition of $f(\phi, u)$. $\therefore f(\phi, u) = v$

$(\Leftarrow) f(\phi, u) = v$. Then, $V_M(\phi, v) = 1 \therefore v$ is a ϕ -world

Also, $\forall u \in U$ st. $V_M(\phi, u) = 1, v \leq_w u$

$\Rightarrow$ for all ϕ -worlds $u, v \leq_w u$

so v is the maximally close ϕ -world to w

$\swarrow \quad \searrow$ This is the same statement?
Pf. Since $V_M(\psi, v) = 1$ iff $V_M(\psi, v) = 1$,

by lemma,

$$V_M(\psi, v) = 1 \text{ iff } V_M(\psi, f(\phi, u)) = 1$$

$$\therefore V_M(\phi \Box \Rightarrow \psi, w) = 1 \text{ iff } V_M(\phi, u) = 0, \forall u \in U$$

$$\text{or } V_M(\psi, f(\phi, w)) = 1$$

(b) Claim: $\models_{sc} (\phi \Rightarrow (\psi \vee \chi)) \rightarrow (\phi \Rightarrow \psi \vee \phi \Rightarrow \chi)$

Assume for reductio that

$$V_M((\phi \Rightarrow (\psi \vee \chi)) \Rightarrow (\phi \Rightarrow \psi \vee \phi \Rightarrow \chi), u) = 0$$

$$\therefore V_M(\phi \Rightarrow (\psi \vee \chi), u) = 1 \quad \&$$

①

$$V_M(\phi \Rightarrow \psi \vee \phi \Rightarrow \chi, u) = 0$$

②

From ①: Either $\forall u \in W, V_M(\phi, u) = 0$
or $V_M(\psi \vee \chi, f(\phi, u)) = 1$

$$(a_k \mid: \forall u \in W, V_M(\phi, u) = 0$$

$$\text{From ②: } V_M(\phi \Rightarrow \psi, u) = 0 \quad \&$$

$$V_M(\phi \Rightarrow \chi, u) = 0$$

$$\therefore V_M(\phi, u) = 1$$

$$\therefore V_M(\phi, u) = 0 \quad \times$$

$$\text{Case 2: } V_M(\psi \vee \chi, f(\phi, u)) = 1$$

$$\text{suppose } f(\phi, u) = v$$

By anti-symmetry of SC, we know that v is unique
~~And that is why f is a~~
 function

$$\therefore V_M(\phi, v) = 1,$$

$$\forall u \in W \text{ s.t. } V_M(\phi, u) = 1, v \leq u$$

$$\text{from (2): } V_M(\phi \supset \psi, u) = 0 \quad \& \quad$$

$$V_M(\phi \supset \chi, u) = 0$$

$$\therefore V_M(\psi, f(\phi, u)) = 0$$

$$\therefore V_M(\chi, f(\phi, u)) = 0$$

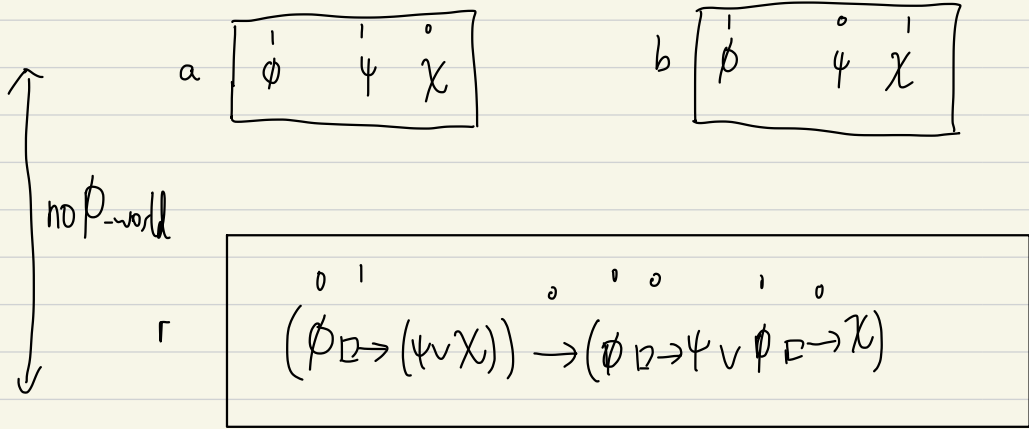
Since if $f(\phi, u) = u$, $f(\phi, u) = v$, then $u = v$

$$\text{we know for } f(\psi \vee \chi, f(\phi, u)) = 1$$

$$\Rightarrow f(\psi, f(\phi, u)) = 1 \text{ or } f(\chi, f(\phi, u)) = 1 \quad \checkmark$$

✓

$$(c) \quad \not\models_{LC} (\phi \supset (\psi \vee \chi)) \rightarrow (\phi \supset \psi \vee \phi \supset \chi)$$



$$W = \{r, a, b\}$$

$$\phi = p, \quad \psi = R, \quad \chi = Q$$

$$I(p, a) = 1, \quad I(p, b) = 1, \quad I(p, r) = 0$$

$$I(R, a) = 1, \quad I(R, b) = 0$$

$$I(Q, a) = 0, \quad I(Q, b) = 1$$

$$\prec = \{ \langle r, r \rangle, \langle b, b \rangle, \langle a, a \rangle, \langle r, a \rangle, \langle r, b \rangle, \langle a, b \rangle, \langle b, a \rangle \}$$

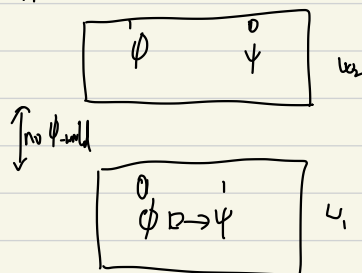
$$\downarrow$$

$$Is this \preceq_r ?$$

✓

Q2 (a) : $\psi \not\models_{sc} \phi \Rightarrow \psi$

Countermodel



let $\phi = p, \psi = r$

$W = \{u_1, u_2\}$

$I(p, u_1) = 0, I(p, u_2) = 1$

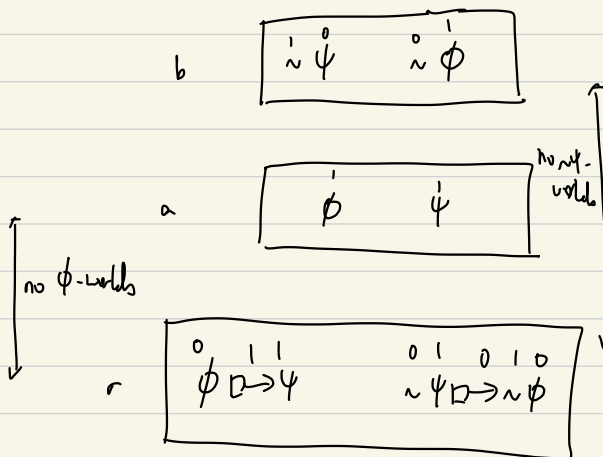
$I(r, u_1) = 1, I(r, u_2) = 0$

$\leq = \{ \langle u_1, u_1 \rangle, \langle u_1, u_2 \rangle \}$

↓ specify which worlds

✓

ii. $\phi \Rightarrow \psi \not\models_{sc} \sim \psi \Rightarrow \sim \phi$



let $\phi = p, \psi = r$

$W = \{r, a, b\}$

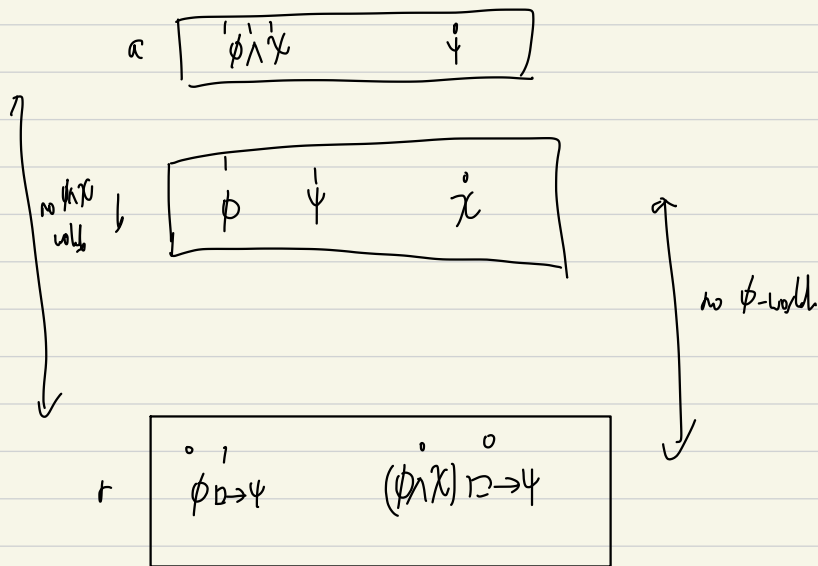
$\leq_r = \{ \langle a, b \rangle, \dots \}$

$I(p, a) = 1, I(p, b) = 1, I(p, r) = 0$

$I(r, a) = 1, I(r, b) = 0, I(r, r) = 1$

✓

$$ii: \phi \Box \rightarrow \psi \not\models_{SC} (\phi \wedge \chi) \Box \rightarrow \psi$$



$$U = \{i, a, b\}$$

$$\angle_r = \{\langle b, c \rangle, \dots\}$$

$$\text{let } \phi = P, \chi = Q, \psi = R$$

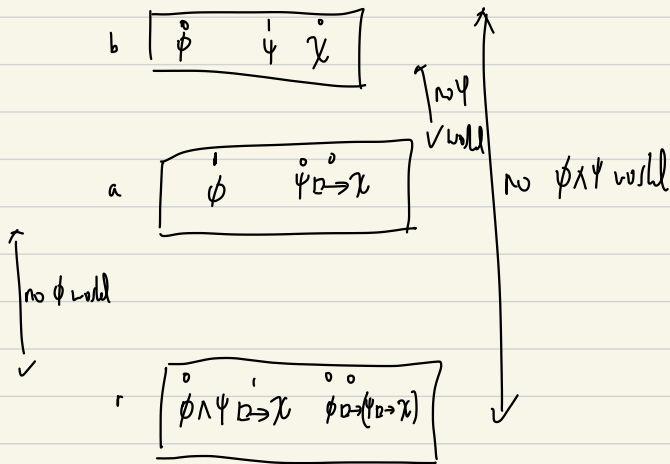
$$I(P, a) = I(Q, a) = I(P, b) = I(R, b) = 1, \text{ all else } 0$$

$$I(P \Box \rightarrow R, r) = 0 \text{ hence } I(P, a) = 1 \text{ and}$$

$$I(R, a) = 0$$

$$\text{Your model works if } \chi = R \text{ and } \psi = Q$$

$$iv. (\phi \wedge \psi) \Box \rightarrow \chi \not\models_{sc} (\phi \Box \rightarrow (\psi \Box \rightarrow \chi))$$



$$W = \{r, a, b\}, \quad \leq_r = \{ \langle a, b \rangle \}$$

let $\phi = P, \psi = R, \chi = Q$

$$I(P, r) = 0, \quad I(P, a) = 1, \quad I(P, b) = 0$$

$$I(R, r) = 0, \quad I(R, a) = 0, \quad I(R, b) = 1$$

$$I(Q, r) = 0, \quad I(Q, a) = 0, \quad I(Q, b) = 0$$

you need to specify $\leq_a$ for this one to show that $V_a(\psi \Box \rightarrow \chi, a) = 0$, but something like $\leq_a = \{ \langle a, r \rangle \}$ should work

(b) (i) $\psi \not\models_{sc} \phi \Box \Rightarrow \psi$ preserves truth for the English counterfactual conditional

Just because say, 'Jane is not pregnant' does not license the inference that

'if diamonds were not shiny, then Jane would have not been pregnant' ✓

because it is completely possible that should the antecedent obtain

in some other world, Jane would have been pregnant in some world that is not in ours

(ii) $\phi \Box \Rightarrow \psi \not\models_{sc} \sim \psi \Box \Rightarrow \sim \phi$ also preserves-truth ✓

'if it rained, it would not rain heavily'
clearly does not license the inference that

'if it does ~~not~~ rain heavily, it would not rain'

$\Rightarrow$ contraposition fails for English counterfactuals

(iii) $\phi \Box \Rightarrow \psi \not\models_{sc} (\phi \wedge \psi) \Box \Rightarrow \psi$

English counterfactual conditionals also do not license augmentation of antecedent ✓

'If I switched the lightbulb on, the light would be on' does not

license 'If I switched the lightbulb on & removed it, the light would be on'

$\Rightarrow$ but, I might add that 'removing lightbulb' entails $\sim$ 'the light would be on'

$\Rightarrow$ more a feature of expressive limitations of MPL

(iv) $(\phi \wedge \psi) \Box \Rightarrow \chi \not\models_{sc} (\phi \Box \Rightarrow (\psi \Box \Rightarrow \chi))$ $\Rightarrow$ preserves-truth in English counterfactuals

✓ English inference clearly does not license the move from 'if I had used red & blue paint, I would have people' to 'if I had used blue, then if I had used red, I would have people' since it might be that

when I used red, I might have used red instead of blue, and thus not obtain purple $\Rightarrow$ it is not a logical validity in some closest world when I use red

Q3. P : I flipped the coin

R : The coin lands on head

Q : The coin lands on tails

Claim: $P \Box \rightarrow R \vee Q, \sim(P \Box \rightarrow R) \models_{SC} P \Box \rightarrow Q$

Pf. Assume for reductio that

$$(1) V_M(P \Box \rightarrow R \vee Q, u) = 1$$

$$(2) V_M(\sim(P \Box \rightarrow R), u) = 1$$

$$(3) V_M(P \Box \rightarrow Q, u) = 0$$

From (1), for a closest P -world, v

$$V_M(P, v) = 1, V_M(R \vee Q, v) = 1, \text{ or } \forall v' \in U, V_M(P, v') = 0$$

From (2), $V_M(P \Box \rightarrow R, u) = 0$

for a closest P -world, u

$$V_M(P, u) = 1 \ \& \ V_M(R, u) = 0$$

From (3), for some closest P -world, u'

$$V_M(P, u') = 1 \ \& \ V_M(Q, u') = 0$$

Given that $\exists u \in W$ s.t. $V_M(P, u) = 1$ from (2),

it cannot be that $\forall u' \in U, V_M(P, u') = 0$

$\therefore$ for a closest P -world, v , $V_M(R \vee Q, v) = 1$

But from (2), (3): $V_M(R, u) = 0, V_M(Q, u') = 0$

We know that for an SC selection function, $f(\phi, u)$

if $f(\phi, u) = u', f(\phi, u) = v$ & $f(\phi, w) = u$, then $u = u' = v$

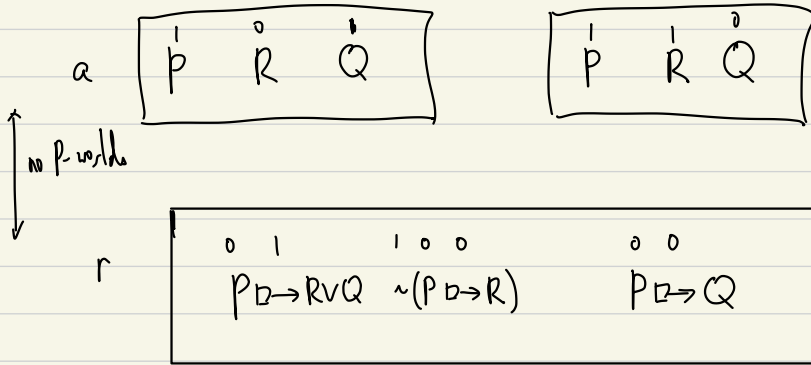
due to limit & anti-symmetry

$\therefore V_M(R, v) = 0, V_M(Q, v) = 0$

$\therefore V_M(R \vee Q, v) = 0$ ✗

✓

$$(v) \quad P \supset R \vee Q, \sim(P \supset R) \not\vdash_{LC} P \supset Q$$



$$W = \{r, a, b\}$$

$$\leq_r = \{\langle a, b \rangle, \langle b, a \rangle, \dots\}$$

$$I(P, r) = 0, I(P, a) = 1, I(P, b) = 1$$

$$I(R, a) = 0, I(R, b) = 1.$$

$$I(Q, a) = 1, I(Q, b) = 0$$

root $\circ$



(c) The conclusion that if I had flipped the coin, it would land on tails does not intuitively follow from the premise, since even if it is the case that flipping the coin does not definitely lead to the coin landing on heads, there is still some possibility that it could land on heads (just not definitely).

So, it is not the case that flipping the coin leads to it landing on tails.